

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(vi)		moles $\text{Ag}^+ = \frac{0.100 \times 26.40}{1000} = 2.64 \times 10^{-3}$ (1) moles Cl^- in 25.0 cm^3 diluted sample = 2.64×10^{-3} moles Cl^- in 25.0 cm^3 undiluted sample = $2.64 \times 10^{-3} \times 5 = 0.0132$ (1) moles Cl^- in 1000 cm^3 undiluted sample = $0.0132 \times 40 = 0.528$ (1) mass Cl^- in 1000 cm^3 seawater = $0.528 \times 35.5 = 18.7 \text{ g}$ (1) answer must be to 3 sig figs		4		4	3	
	(b)	(i)		pour <u>seawater</u> into a beaker and <u>add barium nitrate solution</u> (and allow precipitate to settle) (1) add more drops of barium nitrate (to the clear solution) until no further precipitate forms (1) filter precipitate (making sure that all of it is transferred to the filter) (1) wash precipitate and dry (1) award (1) for either of following • heat to constant mass • subtract mass of filter paper from mass of precipitate and filter paper	5			5		5
		(ii)		moles $\text{BaSO}_4 = \frac{0.65}{233.1} = 2.789 \times 10^{-3}$ (1) moles $\text{Ba}(\text{NO}_3)_2 = 2.789 \times 10^{-3}$ volume $\text{Ba}(\text{NO}_3)_2 = \frac{2.789 \times 10^{-3}}{0.100} = 0.02789 \text{ dm}^3$ (1) volume = $27.89 / 27.9 / 28 \text{ cm}^3$ (1)		3		3	2	